MILITARY SPECIFICATION

COMPOSITIONS, TRACER, R-45 AND IGNITER, K-29

1. SCOPE

1.1 This specification covers igniter composition, K29 and tracer composition, R-45, intended primarily for use in loading of artillery ammunition to obtain a visible trace.

2. APPLICABLE DOCUMENTS

2.1 The following specifications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard For Testing Purposes

MILITARY

MIL-B-153 - Barium Peroxide
JAN-G-155 - Graphite (For Use in Ammunition)
JAN-M-382 - Magnesium Powder (For Use in Ammunition)
JAN-M-454 - Magnesium-Aluminum Alloy, Powdered
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition
MIL-P-20307- Polyvinyl Chloride (For Use in Pyrotechnics)
MIL-S-20322- Strontium Nitrate, Non-Hygroscopic
MIL-C-20470- Calcium Resinate

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The materials used in the tracer and ignition compositions shall conform to the requirements specified hereinafter.

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3.1.1 Strontium nitrate.-The strontium nitrate used shall be anhydrous and shall comply with the requirements of Specification MIL-S-20322, Grade B.

3.1.2 Magnesium-aluminum alloy, powdered.-The powdered magnesium-aluminum alloy shall comply with the requirements of Specification JAN-M-454, Type B, except that it shall comply with the following granulation requirements using U. S. Standard sieves complying with the requirements of Specification RR-S-366:

Through No. 100 sieve - 85.0 percent, minimum
Through No. 200 sieve - 15.0 percent, minimum

3.1.3 Barium peroxide.-The barium peroxide used shall be chemically pure, minimum 88 percent BaO_2, and shall comply with the requirements of Specification MIL-B-153, Class C.

3.1.4 Magnesium powder.-The magnesium powder used shall comply with the requirements of Specification JAN-M-382, Type III, Granulation 15.

3.1.5 Calcium resinate.-The calcium resinate used shall comply with the requirements of Specification MIL-C-20470, either Type I or Type II.

3.1.6 Graphite.-The graphite used shall comply with the requirements of Specification JAN-G-155.

3.1.7 Polyvinyl chloride.-The polyvinyl chloride shall comply with the requirements of Specification MIL-P-20307.

3.2 Composition.

3.2.1 Tracer composition, R-45.-The tracer composition, R-45, shall be as follows, when determined as specified in 4.4.2.

Component	Analysis of finished mixture (percent)
Strontium nitrate, anhydrous	56.0 ± 2.0
Magnesium-aluminum alloy, powdered	37.0 ± 2.0
Polyvinyl chloride	7.0 ± 1.0

3.2.2 Igniter composition, K-29.-The igniter composition, K-29, shall be as follows, when tested as specified in 4.5.2 (see 6.3):

Component	Analysis of finished mixture (percent)
Barium peroxide (88 percent purity)	80.5 ± 2.0
Magnesium (free metallic) powder	16.5 ± 2.0
Calcium Resinate	2.0 ± 0.5
Graphite	1.0 ± 0.5

3.3 Moisture content.-The moisture content of either composition shall not exceed 0.10 percent when tested as specified in 4.4.1 or 4,5.1, as applicable (see 6.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material from the same batch or blending operation, subjected to the same processing operations and conditions.

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4.2 Sampling.-Twenty primary samples shall be selected from each lot of material and subjected to acceptance tests as specified herein. Not more than one primary sample shall be selected from each batch. If there are fewer than 20 batches in each lot, each batch shall be sampled. If no defective batches are found, the lot shall be considered acceptable. If one or more defective batches are found the lot shall be rejected subject to correction and resubmission for test.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Test of tracer composition, R-45.-The moisture content of the tracer composition shall be determined and the ingredients shall be analyzed as follows:

4.4.1 Moisture content.-An accurately weighed portion of approximately 5 grams of the sample shall be transferred to a tared moisture dish 2 inches in diameter. The dish and contents shall be placed in an oven at 135° ± 2°C for 2 hours, cooled in a desiccator and weighed. The loss in weight of the dish and contents shall be calculated to percent of moisture in the sample.

4.4.2 Composition.

4.4.2.1 Polyvinyl chloride.-An accurately weighed portion of approximately 1 gm of the sample material shall be transferred to a 250-ml beaker. Approximately 100 ml of reagent grade methyl ethyl ketone shall be added to the beaker, the beaker covered with a watch glass, and transferred to a steam plate. The beaker shall be heated for one hour with occasional stirring, and then removed from the steam bath. The supernatant liquid shall be decanted through a tared medium-porosity sintered glass crucible set up in a vacuum flask and the residue washed quantitatively into the same crucible using approximately 50 ml of methyl ethyl ketone. The crucible and contents shall be aspirated until the odor of methyl ethyl ketone is no longer detectable. The crucible and contents shall be dried for one-half hour in an oven maintained at 105° ± 5°C, then cooled in a desiccator and weighed. The loss in weight of the crucible and contents shall be calculated to percent polyvinyl chloride, as follows:

$$\text{Percent polyvinyl chloride} = \frac{[(A - B) - (WM)]}{W(1-M)} \times 100$$

where:

W = weight of the sample
M = percent moisture expressed as a decimal fraction
A = weight of the crucible plus weight of the sample
B = weight of the crucible plus weight of the sample, after extraction with methyl ethyl ketone.

4.4.2.2 Strontium nitrate.-The residue obtained in the above determination of polyvinyl chloride shall be leached with successive 10-ml portions of water and cooled to approximately 10°C, until the strontium nitrate has been extracted. A drop of 1-percent diphenylamine solution in 36N sulfuric acid shall be added to 1 drop of the filtrate on the identation of a white spot plate to determine whether the strontium nitrate has been extracted. The absence of a blue coloration shall indicate the complete removal of the strontium nitrate. The crucible and contents shall be placed in an oven maintained at 135° ± 2°C for 1 hour, cooled in a dessiccator and weighed. The percentage of strontium nitrate in the sample shall be calculated on a dry basis, as follows:

$$\text{Percent strontium nitrate} = \frac{(A - B)}{W(1-M)} \times 100$$

where:

A = weight of the crucible and residue after extraction with methyl ethyl ketone
B = weight of the crucible and residue after leaching with water
W = weight of the sample
M = percent moisture expressed as a decimal fraction.

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4.4.2.3 Magnesium-aluminum alloy, powdered.-The percentage of powdered magnesium-aluminum alloy in the sample shall be calculated by subtracting from 100 the combined percentages of polyvinyl chloride and strontium nitrate.

4.5 Test of igniter composition, K-29.-The moisture content of the igniter composition shall be determined and the ingredients shall be analyzed as follows:

4.5.1 Moisture content.-An accurately weighed portion of approximately 5 grams of the sample shall be transferred to a tared moisture dish 2 inches in diameter. The dish and contents shall be placed in an oven maintained at 135° ∠ 2°C for 2 hours, cooled in a desiccator and weighed. The loss in weight of the dish and contents shall be calculated to percent moisture in the sample.

4.5.2 Composition.

4.5.2.1 Barium peroxide (calculated from available oxygen.)-An accurately weighed portion of approximately 0.5 gm of the sample (approximately 100 mesh) shall be transferred to a tall form beaker of approximately 300 ml capacity. Fifty milliliters of distilled water (cooled to a temperature of 10° ∠ 2°C) containing two drops of 1% manganese phosphate solution (made by dissolving 100 mg of manganese in 10 ml of approximately 10% phosphoric acid) shall be added. The contents of the beaker shall be titrated with approximately 0.1N potassium permanganate solution, (see 6.5) containing 10% by volume of 85% phosphoric acid. The titration shall be carried out rapidly by adding most of the required permanganate solution as fast as the burette allows and with a little stirring of the solution. Several ml before titration is finished, the contents of the beaker shall be stirred sufficiently to dissolve all the undissolved barium peroxide. The titration shall be continued until the addition of 1 drop of the permanganate solution causes a coloration which is permanent for one minute. The percentage of barium peroxide shall be calculated as follows:

$$\text{Percent barium peroxide} = \frac{8.4685\ NV}{0.88W\ (1-M)}$$

where:

V = volume of permanganate solution used, in ml
N = normality of permanganate solution
W = weight of sample
M = percent moisture expressed as a decimal fraction

4.5.2.2 Magnesium powder.

4.5.2.2.1 Apparatus.-The apparatus shall be assembled as shown in figure 1. The gas evolution apparatus shall consist of a 600 ml Erlenmeyer flask fitted with a 2-hole rubber stopper. A 25-ml burette for 10% sulfuric acid shall be inserted through one hole of the rubber stopper and by means of the other hole, the Erlenmeyer gas evolution flask shall be connected to a "U" shaped drying tube (see 6.6) containing "Ascarite". The drying tube shall be used only to absorb any evolved carbon dioxide. A 3-way "T" stopcock shall be inserted between the gas burette and the drying tube so that the Erlenmeyer flask can be exposed to atmospheric pressure. The gas burette (calibrated from 450 ml to 500 ml) shall be connected to a water leveling bulb (see 6.7) filled with water saturated with hydrogen. The gas burette shall be water-jacketed to achieve accurate and rapid temperature control. A circulating pump shall be used to circulate the water through the water jacket and a pail of water used as a reservoir. The Erlenmeyer flask shall be immersed in the water in the pail to maintain it at the same temperature as the gas burette.

4.5.2.2.2 Procedure.-An accurately weighed portion of approximately 2.4 grams

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of the sample shall be transferred to a tared medium porosity sintered-bottom glass crucible. The contents of the crucible shall be extracted with six 5-ml portions of chloroform. The chloroform shall be allowed to drain through the residue in the crucible before applying suction. After addition of the last portion, the residue shall be air dried by means of suction until the odor of chloroform is no longer discernible. The crucible and contents shall be transferred to a 600-ml Erlenmeyer flask. Two hundred milliliters of distilled water saturated with hydrogen shall be added. The Erlenmeyer flask shall be connected to the apparatus. A 3-way "T" stopcock shall be opened so that the flask and the gas burette are exposed to atmospheric pressure. The zero point level of the gas burette shall be adjusted with the leveling bulb. The stopcock shall be turned so that it will only allow gas to flow from the Erlenmeyer flask to the gas burette. The leveling bulb shall be lowered, and 25 ml of 10% sulfuric acid shall be added from the 25-ml burette. After the addition of the acid, the flask shall be agitated occasionally until the reaction is completed. The reaction shall be considered complete when the volume of generated gas remains within 1 ml for a period of 20 minutes. When the reaction is complete, the level on the gas burette shall be adjusted to that of the water leveling bulb and the volume of gas in the gas burette shall be read. If the evolved gas is less than is measurable by the gas burette, a known volume of water shall be added through the acid burette until a readable volume is reached on the gas burette. The determination shall be started and finished at the same temperature (approximately 25°C). The percentage of magnesium in the sample shall be calculated on a dry basis, as follows:

$$\text{Percent magnesium} = \frac{(A - B)\ (C - D) \times .03901}{W\ (273 + t)\ (1-M)}$$

where:

A = volume read on gas burette, in ml
B = volume of acid or volume of acid plus water added from burette, in ml
C = atmospheric pressure, in mm. (see 6.8)
D = vapor pressure of water, in mm, at the temperature of the determination
t = temperature at which the determination is made in degrees Centigrade.
W = weight of sample, in grams
M = percent of moisture expressed as a decimal fraction.

4.5.2.3 Calcium resinate.-An accurately weighed portion of approximately 1 gram of the sample shall be transferred to a tared medium-porosity sintered glass crucible. The contents of the crucible shall be extracted with six 5-ml portions of chloroform. The contents of the crucible shall be stirred after the addition of each portion and the chloroform allowed to drain through the residue in the crucible before applying suction. The crucible and contents shall be dried in an oven maintained at 135° ± 2°C for 1 hour, cooled in a desiccator, and weighed. Reserve the crucible and contents for the graphite determination. The percentage of calcium resinate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent calcium resinate} = \frac{[(A - B) - MW] \times 100}{W\ (1-M)}$$

where:

A = weight of the crucible and residue before extraction with chloroform.
B = weight of the crucible and residue after extraction with chloroform.
M = percent moisture expressed as a decimal fraction.
W = weight of the sample.

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4.5.2.4 Graphite.- The crucible and contents reserved from the calcium resinate determination shall be transferred to a 250-ml beaker. One hundred fifty milliliters of 5-percent acetic acid solution shall be added, the beaker and contents covered with a watch glass, transferred to a hot plate, and heated to boiling. Any lumps of undissolved barium peroxide shall be broken up with the aid of a glass stirring rod. The beaker and contents shall be removed from the hot plate, and then the filtering crucible shall be carefully removed. Any adhering particles of graphite shall be quantitatively washed back into the beaker with distilled water. The graphite shall be collected by filtering the contents of the beaker while still hot, through the same sintered glass crucible. The graphite shall be washed with three 20-ml portions of 6N hydrochloric acid followed by three 20-ml portions of hot 8N nitric acid. Each portion shall be allowed to remain in contact with the residue in the crucible for 1 minute before applying suction. The crucible and contents shall be washed with three 20-ml portions of distilled water, dried in an oven maintained at 100°C for 1 hour, cooled in a desiccator, and weighed. The percentage of graphite in the sample shall be calculated on a dry basis, as follows:

$$\text{Percent of graphite} = \frac{(A - B)(100)}{W(1-M)}$$

where:

A = weight of the crucible and residue after extraction
B = weight of empty crucible
M = percent moisture expressed as a decimal fraction
W = weight of the sample.

4.6 Resubmission and retest.- The contractor shall have the option of having a partial or complete analysis made on samples taken from each batch in the lot at no expense to the Government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance. On a resubmitted lot 30 samples shall be selected. The lot shall be accepted on resubmission provided each sample passes all the tests required by this specification. If any sample fails to comply with the requirements of this specification the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Not applicable (see 6.9).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.2 Preparation of tracer composition, R-45.- It is recommended that this composition be prepared in a Simpson mixer in 25-pound batches. The dry ingrediants are weighed, placed in a Simpson mixer with the mullers raised about 1/8 inch off

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the bottom of the mixer. A sufficient amount of methyl ethyl ketone is added until a thick paste is formed. The mixer is run until the mix is granular but still wet with methyl ethyl ketone (about 2 hours). The mixture is then dried at 80°C,

6.3 Preparation of igniter composition, K-29.- It is recommended that this composition be prepared in a Lancaster mixer in 5 pound batches. The barium peroxide and magnesium are weighed and placed in the Lancaster mixer with the mullers raised about 1/16 inch above the bottom of the pan. The calcium resinate is dissolved in sufficient 95 percent ethyl alcohol to make 10 percent solution and then added in the mixer. Sufficient additional alcohol is added, if necessary, to make a thick paste. The mixer is run until the mix is granular but not dry (about 2 hours). The mixture is then dried at 80°C.

6.4 These compositions, and the pellets formed thereof, are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in, or absorbed by, the compositions or pellets, prior to or after loading into ammunition, will eventually result in deterioration with resultant lowered candlepower, lowered visibility, split igniters, partial and total blinds. It is highly desirable that the ingredients, compositiins, pellets and loaded ammunition' items be carefully safeguarded from exposure to moisture and humid atmospheres. For assurance of a satisfactory end-product it is highly desirable that the building be air conditioned in a manner which will hold the humidity of the atmosphere as low as practicable.

6.5 The permanganate solution used in making the barium peroxide determination (see 4.5.2.1) may be standardized by any suitable method. However, in case of non-compliance, dispute, or question the standardization should be made using National Bureau of Standards sodium oxalate, following the method described in Scott's "Standard Methods of Chemical Analysis", D. VanNostrand Co., New York, 5th Edition, Volume I, pp. 1209-1210.

6.6 A Schwartz U-shaped glass stoppered drying tube (such as Fisher Catalog No. 9-230) has been found satisfactory for use in making the magnesium content determination (see 4.5.2.2.1).

6.7 A leveling bulb of 1-liter capacity (such as Fisher Catalog No. 10-788) has been found satisfactory for use in making the magnesium content determination (see 4.5.2.2.1).

6.8 In making the magnesium content determination it has been found satisfactory to measure the atmospheric pressure with an aneroid barometer (see 4.5.2.2.2).

6.9 The tracer and igniting compositions and the pellets made thereof are not intended for shipment. They should be prepared at the plant where loading of the ammunition will be accomplished.

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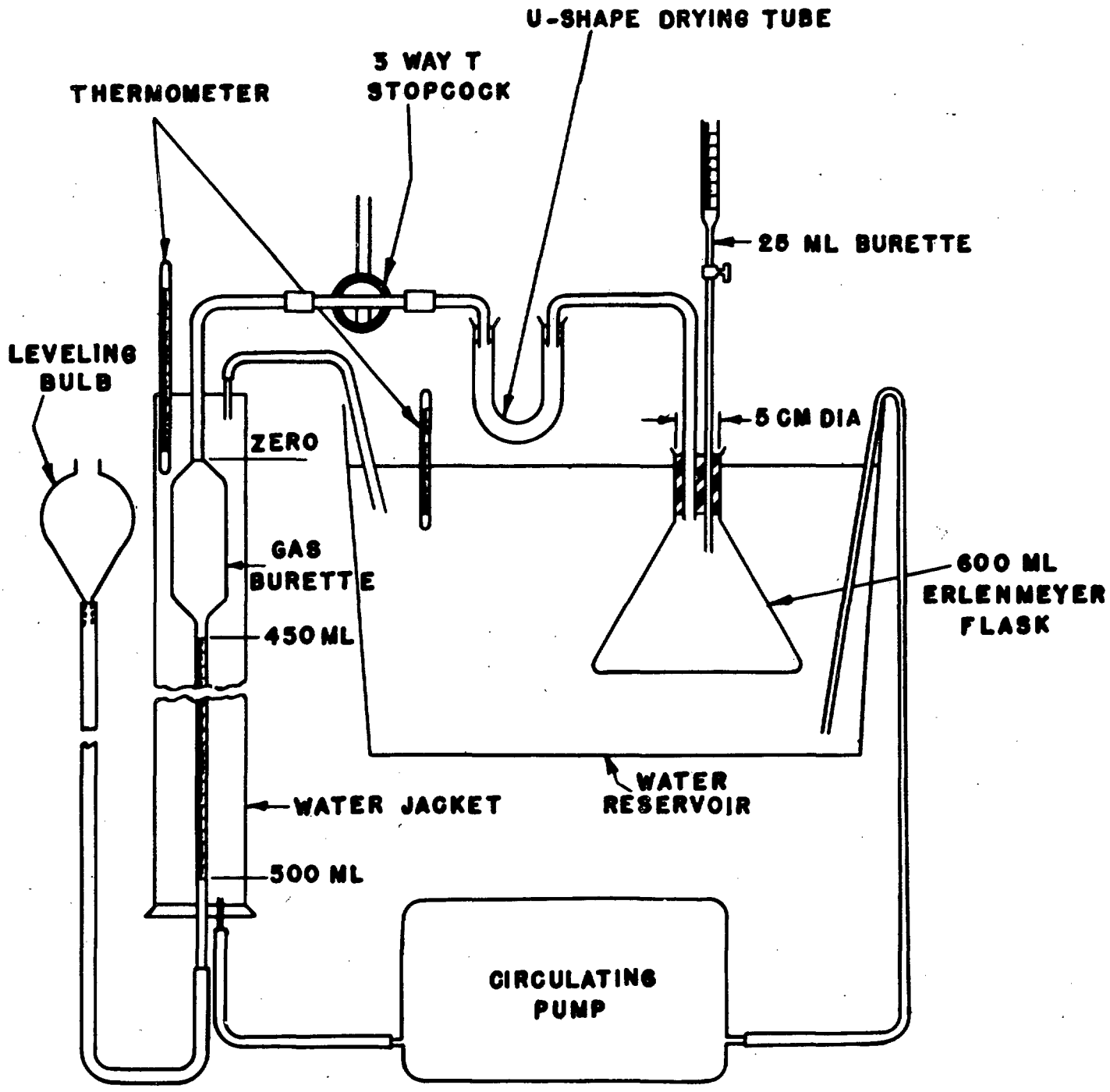


FIG. 1-EQUIPMENT ASSEMBLY

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